

CONFIDENTIAL MEMORANDUM

To: Engineering Team

From: Architecture Group

Date: October 2023

Subject: RAG Pipeline Optimization Strategy

Executive Summary

The current PDF parsing library has proven to be a bottleneck for our ingestion pipeline. Due to the Python Global Interpreter Lock (GIL), processing pages concurrently yields no measurable speedup. We propose writing a Rust core that extracts raw text and routes scanned pages to Tesseract, effectively bypassing Python until the final chunking stage.

Implementation Details

1. Extract digital-native text directly using lopdf.
2. Identify scanned pages (no text objects) and pass them to OCR.
3. Provide section metadata to the downstream chunker to automatically discard headers and footers.

